

What is bioinformatics?

1

1.1 Introduction

Bioinformatics is an interdisciplinary life science field. This is a field which deals with the collection and efficient analysis of biological data. In other words, it is a recently developed science which uses information to understand biological phenomenon. The research in bioinformatics is regarded as a domain which encompasses expanding, complex, and large datasets. It is a part of computational biology which addresses the necessity of managing and interpreting the data, massively generated in the past decade by genomic research. Bioinformatics is the discipline that integrates the biotechnology and information technology, interpretation and analysis of data, genomics convergence, development of algorithm and modeling of biological phenomena. Bioinformatics is a wide encompassing branch and is therefore difficult to define. For some it is still an ambiguous term encompassing biological modeling, system biology, biophysics and molecular evolution. Whereas for others it is simply computational science applied to a biological system.

Bioinformatics entails the usage of high technology solutions in biological experiments through a variety of computer programs. Bioinformatics is becoming a vital part of biology. Bioinformatics uses image and signal processing technology to derive essential knowledge from vast volumes of data. In genetics, genome sequencing and mutation analysis may be significantly important. Information plays a vital role in the research for biologists and the creation of gene ontologies. It plays an essential function in the study of gene and protein influence. Bioinformatics tools help in explaining and contrasting molecular biology evidences. It helps in recognizing the biological processes and biological networks that are active in the biological system. We need this in structural biology to simulate and forecast molecular activity.

Bioinformatics is a proliferating area which is currently in the foreground of science and technology. Various institutes all over the world are heavily investing (especially because of the pandemic) in possessing, transferring and exploiting the data for future development. It is a valuable ticket at present, and bioinformatics learners would thrive from the demand for jobs in private sector, in government and academia.

There are several elements of science that contribute to bioinformatics. It also implies to biological molecules and thus includes knowledge of the fields of

molecular engineering, molecular biology, statistical mechanics, biochemistry, thermodynamics, molecular evolution, and biophysics. The use of computer science, mathematical, and statistical principles are needed in the field. Bioinformatics is at the intersection between experimental and theoretical research. Its not just about “mining” data or modeling, it is about analyzing the molecular environment that drives life from the perspective of evolution and mechanisms. It is genuinely cross-disciplinary and is evolving. Like genomics and biotechnology, bioinformatics is evolving from applied to fundamental research, from creating tools to creating hypotheses.

Bioinformatics, Computational biology, and bio information infrastructure are sometimes used interchangeably.

1. **Bioinformatics** relates to the methods, observations, and data storage utilized in the genomic era.
2. **Computational biology** requires the usage of software to analyze biological processes better.
3. **Bio information infrastructure** includes all the information software, computational methods, and networking networks supporting biology. The latter dissertation may be seen as an informational scaffold for the first two.

1.2 History

Bioinformatics was first properly established 50 years back. Although the term “Bioinformatics” was coined by Ben Hesper and Paulien Hogeweg in 1970 ([Hesper and Hogeweg, 1970](#)), but the tracks of its emergence go back to 1960s with the efforts put by Margaret Oakley Dayhoff, Russell F. Doolittle and Walter M. Fitch ([Chang et al., 1965](#)). The contribution of Margaret is so important to this field that former director of NCBI David J. Lipman called her “the mother and father of bioinformatics”. There was need to compare and analyze a huge amount of protein sequences or amino acids sequences from different organisms computationally as it was manually impractical to handle such large data. This is what lead Margret O. Dayhoff, “the first Bioinformatician” and her colleagues at the National Biomedical Research Foundation in compiling the first ever “Protein Information Resource” (PIR), stating that “analysis of protein was the starting point for bioinformatics” ([Dayhoff et al., 1974](#)). They successfully organized the protein sequence data into various groups and subgroups according to the requirement. Further contribution to the development of bioinformatics was given by Elvin A. Kabat in the 1970s by his extended analysis of protein sequences of comprehensive volumes of antibody sequences. Further in 1974, George Bell and his associates initiated the DNA sequences collection into the GenBank, with the objective of contributing to the theoretical background to immunology. The primary version of GenBank was being prepared in 1982–1992 by Walter Goad’s group ([Burks et al., 1987](#)). Consequently, The DNA Data Bank of Japan (DDBJ) and European Molecular Biology Laboratory

(EMBL), “the world’s first nucleotide sequence database” were also made in 1984 and 1980 respectively.

The first conceptualization of Bioinformatics in Switzerland was in the early 1980s. Swiss bioinformaticians developed software to compare genetic nucleotide sequences, created programs for the study of experimental peptide and protein results, invented computer tools for three-dimensional modeling structures of proteins, and created databases of protein details. These individuals participated in the field of bioinformatics and contributed to biology science in general. In 1998, Swiss bioinformatics became unified. The current five Swiss bioinformatics groups combined to create the SIB (Swiss Institute of Bioinformatics), a charitable organization.

However, the most important development in these databases was the incorporation of the web-based searching algorithms which helped researchers in their queries. GENEINFO was the resulting computer software developed by David Benson, Lipman and associates. The software was made available through NCBI (National Center of Biotechnology Information) web-based interface. Also, NCBI was made available online in 1994 along with the tool BLAST (Altschul et al., 1990). Afterward, several major databases which are still in usage like PubMed (1997) and Human Genome (1999) came into existence.

Bioinformatics tools are growing evermore prolific and are increasingly expected to replicate all results. To help students understand evolution more accurately, professors are integrating this theory into biology students’ curriculum. Synthetic biology, systems biology, and whole-cell modeling have emerged due to the ever-increasing complementarity between computer science and biology (Hagen, 2000).

1.3 Biological databases

A biological database is a complex, extensive and complete structured collection of biological data arranged in computer readable form which enhances the search speed and retrieval. Biological databases have appeared as a response to the massive amount of data provided by the low-cost DNA sequencing technologies. The first database to develop was GenBank, which is a compilation of all the accessible DNA and cDNA sequences.

Previously databases were perceived somewhat different. However, over the course of time the term “biological database” has become a default concept. The data is directly submitted to the biological databases for organization, indexing and optimizing the data. The databases help students, scientist and researchers to find, discover and analyze the related biological data by making it accessible in a format which can be read and used on software’s. This is the primary purpose of these databases, storing, managing and retrieval of biological information. A range of information can be retrieved from these biological databases like binding sites, molecular actions, biological sequences, metabolic interactions, motifs, protein families, molecular action, homologous and functional relationships etc. A lot of Bioinformatics work is based on data collection and manipulation. Any of these provide both “public” and

“private” database references. Having those databases available to different computers allows it far simpler for more users to communicate these databases efficiently.

Biological databases may be broadly classified into PRIMARY, SECONDARY and DERIVED databases. And then there are further distribution. Primary databases contain only sequential and structural information. These can also be called as archival databases. They are loaded with experimentally generated data such as protein and nucleotide sequences. Experimental data are directly submitted to database by scientist or the researcher. After the database accession number is given, the primary database data will never be changed: they become the part of the scientific record. Few examples of primary biological databases are: GenBank and DDBJ for genome sequences, EMBL, Swiss-Prot and PIR for protein sequences and Protein Databank for protein structures.

Secondary databases are those which constitutes data/information from primary databases or the analyzed result of the primary databases. The primary databases often have minimal sequence annotation information. A much more post processing of the sequence data is required to convert raw data of sequences into more sophisticated biological information. This implies the necessity for databases containing computationally analyzed sequencing data which is obtained from primary databases. Hence secondary database comes into the picture containing information of the results of the analysis of primary data. Secondary databases are highly curated and consists of more valuable information in comparison to the primary databases. The databases comprise data such as signature sequence, active site residues and conserved sequences. Few examples of secondary databases are: UniProt KB, Motif databases, PDB and InterPro etc. (Fig. 1.1).

Composite or derived databases are the amalgam of the primary and secondary databases. To enter the data as input in the database, it is first compared and then sorted on the basis of the desired parameters. Primary databases are source for extracting initial data and is then combined in conjunction based on specific parameters. They consist non-redundant data. Examples of composite databases are: OMIM (Online Mendelian Inheritance in Men) and Swissport. Also, there are databases which are specialized for a particular research interest, for e.g. HIV sequence database, Flybase and Ribosomal Database project.

1.4 Algorithms in computational biology

Computational biology/Bioinformatics are interdisciplinary areas which are concerned with employing computers capacities to address biological interests' problems. The two terms are used interchangeably, but there's a consensus formed between the two. Bioinformatics refers or focuses on the activities which gives attention on developing and utilizing computational tools for the analyses of the biological data. Whereas Computational biology refers to those activities which mainly works on constructing or developing algorithm leading to address the biologically relevant problems.

Biological Data

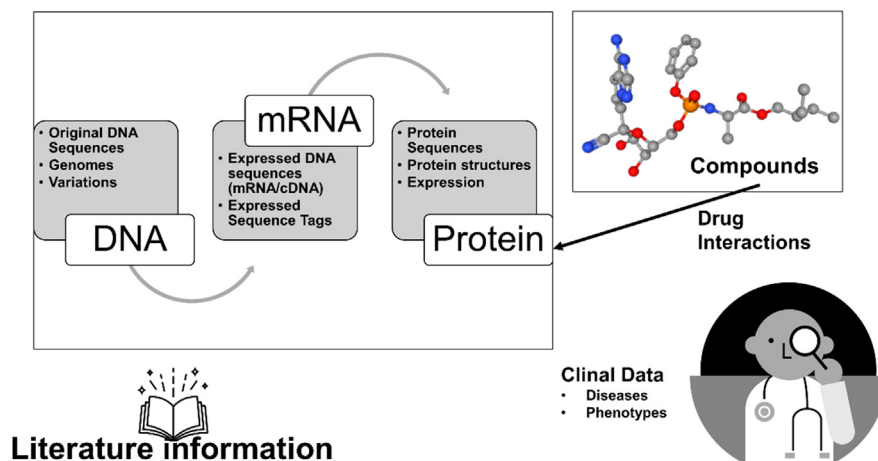


FIGURE 1.1

Biological information to data.

Up until recently biologists have not had access to such massive quantity and quality of data generated and stored in different databases discussed above. Over the past 2 decades, unprecedented technical advancement has been made in producing biological data, techniques like microscopy, next generation sequencing, high throughput techniques etc have contributed to data explosions. Researchers are producing datasets that are so enormous that it has become impossible to analyze, manage and make proper use of the data to understand biological process and their relationships. This is what led to the introduction of the various algorithms in the field. An algorithm is a process or description about how to solve a problem. Modern day computer's ability to perform and store billions of calculation and processes makes it possible to use the amount of data generated not only in just biology but any other field. The computational biology algorithms have several uses including prove or disapprove a certain hypothesis.

The process of creating algorithms that resolves biological significant issues, in computational biology comprises of two steps. First phase is to raise an interesting biological question and to build a model of biological reality that makes it possible to articulate the question as a computational problem. Secondly, construct an algo which will be able to solve the formulated computational problem. The primary step needs biological reality knowledge, while the latter requires algorithmic theory knowledge. The algorithm quality is a combination of its space assumption and running time and the answers of biological relevance it produced. Data scientist frequently torture the various data structure in order to reduce the ambiguity of space and time. This approach has explicitly benefited researchers well in the field. Hence

having a working knowledge of basic computational algorithm is of paramount importance to bioinformaticians or researchers in the field, also their expertise in the development of novel algorithm would have a strategic advantage in both academia and industry (Fig. 1.2).

There are many algorithms already existing in the field, helping in the current research. Some of them are Dynamic programming: Needleman Wunsch (Global alignment) and Smith waterman (Local alignment), Hidden Markov Models, Principal Component Analysis Clustering, Phylogenetic tree construction, machine learning applications (SVM, neural network), microarray data analysis, protein secondary structure prediction and many more.

Global and Local sequence alignment uses our understanding of a organism's proteins to understand more about the proteins of other organisms. Next HMM are used for sequence modeling or model a DNA sequence. In HMM, the probability of happening of an event is dependent on its previous state. This model uses a probabilistic finite state machine in which a letter is emitted depending upon the present state probability and then move to next state. The next state can be possibly equal to the original one. In gene regulation networks, they formed because of the different protein's interaction in an organism. The various proteins regulate each other and, depending on the structure of their interactions, the cell type is determined (Crombach and Hogeweg, 2008).

1.5 Genetic variation and bioinformatics

Genetic variations are the modifications (changes) in the chromosome sequences. Variation is also the reason why two individuals of the same species having similar characteristics, but are not identical. It is the engine of evolution which enables

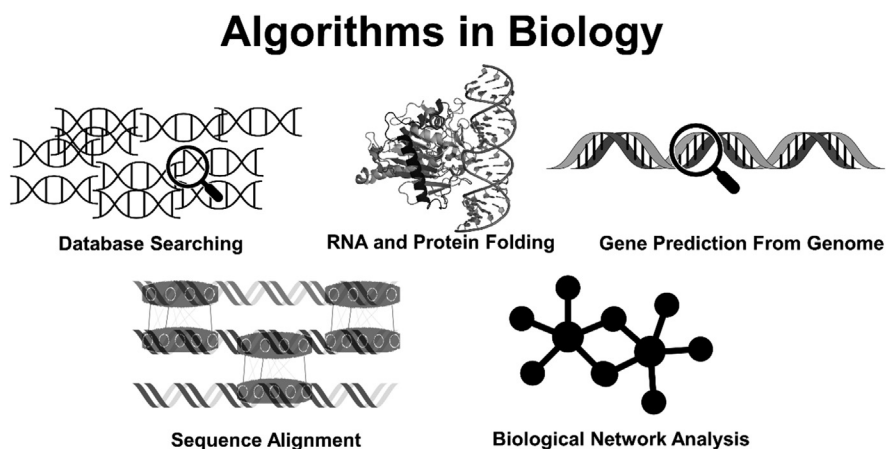


FIGURE 1.2

Challenges in biology solved through algorithms.

organisms to conquer the environmental challenges they meet (Stoletzki, 2008). It may be both damaging and effective in the development of efficient mechanisms in cell factories to tackle changes and survive. In the evolution of biotechnology, efforts have been made to make use of genetic variation to our advantage in order to produce strains with beneficial phenotype. It is also stated as the variation in the sequences of DNA among people within a population and it happens in somatic cells as well as germ cell (egg and sperm cells). The only difference that exists is the variation in germ cell can be inherited through one person to another person, hence impacting population dynamics and subsequently evolution. The main cause of variance are recombination and mutations. Mutations are said to be the original source of variance causing permanent alteration in DNA sequence. It can be harmful, beneficial or neutral to the organism. The other main reason for genetic variation is recombination. Every organism has a combination of genetic information from their parents. Thus, recombination happens when these genetic materials combine or say homologous DNA strands are crossed and aligned. SNPs (single nucleotide polymorphism) are the genetic variation which is very common among people. Every SNP reflects a variation in the DNA base A, G, C, T of a person's genome. They occur on average once in every 300 bases and are also present between genes. The core priority area of modern medical research is studying the effect of SNPs on human health. In more than 1% of population Single nucleotide substitution can be observed. Numerous algorithms have been applied to evaluate the impact of SNPs mainly focused on the human genotype data analysis which classifies variations either diseases-causing or neutral, tolerant or intolerant and deleterious or neutral. Which implies that the genetic variation would either expected to have no impact or inflict some significant negative effects on the phenotype. The one downside to these algorithms is that they are classifiers build on existing knowledge and it is well said that biology is the science of exceptions, presently scientific community has able to uncover only the tip of the ice burg representing biological phenomena. Therefore, these tools are built on the assumptions which we have onboard presently. These tools are used for predicting disorders and are mainly used for diagnostic purposes. There are many available tools and databases for predicting the effects of SNPs. Some of them are Variant Effect Predictor (VEP) which evaluates the impact of variants on genes, protein sequences, transcript etc., SIFT (Sorting Intolerant from Tolerant) is sequencing homology-based tool which filters tolerant amino acids and also determines whether the substitution of amino acid in protein would have a phenotypic effect. DbSNP is the SNP database of NCBI, which contain information on non-polymorphic, microsatellite and deletion/insertion forms.

Bioinformatics in Genetic variation covers the following areas;

- (a) Latest algorithms and software development for genetic variance analysis and application with pipelines and visualization tools.
- (b) Genetic variations analysis in the genome; DNA and single nucleotide polymorphisms (SNPs); techniques to assess numbers of people with a disease; study large-scale data sets.

- (c) It involves studies of data sets, and study of recent methodological advances in the area of genetics.
- (d) Involve in genetic variance identification, functional annotation, pathway simulation, and analytical methods built for different sequencing platforms.

1.6 Structural bioinformatics

“Structural bioinformatics is a subset of Bioinformatics that deals with the prediction and analysis of 3D structure of Macromolecules such as DNA, RNA, and Proteins.” And the second thing which comes is why understanding the structure of macromolecule is important. The reasons are first: structure determines function, so learning structure helps in understanding of function. Secondly the Structure is more conserved than the sequence, hence enabling identification of a much more distant evolutionary relationship. Thirdly understanding the structural determinants enables the design and modification of proteins for industrial and medical benefit. The structural bioinformatics field and concepts related to it offers not only a way of coordinating views about sequence-structure-function questions but also a mechanism for detecting unobserved behavior and proposing novel experiments (Konings et al., 1987; Schuster et al., 1994).

Proteins are essential components of cells of living species. The structural specificity of a protein is related to the role of the protein. Protein structure visualization is a subject of recent biochemistry research and is an essential method for structural bioinformatics. Most often used is: **Cartoon**: This illustrates the secondary structure variations for the protein. Besides, as α -helix is often interpreted as a form of a screw, β -strands is also described as arrows, and loops as arcs. **Lines**: Each atom is depicted by thin lines and allows for a lower cost of data in a visualization. **Surface**: In this visualization, one can see how the molecule appears. **Sticks**: There are covalent connections between amino acid atoms in proteins. This kind of cluster graph strategy is most widely used for visualizing relationships between amino acids (Fig. 1.3).

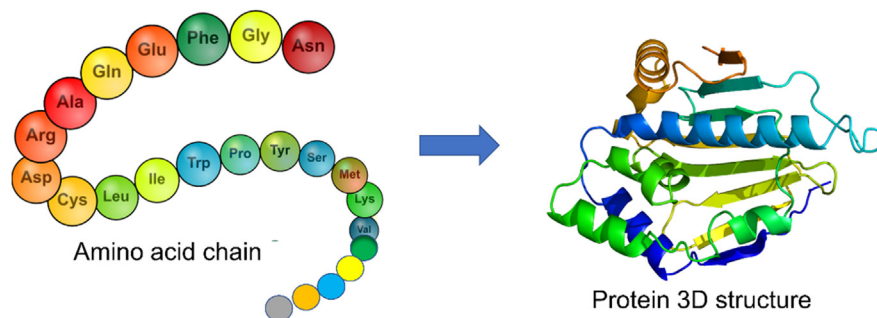


FIGURE 1.3

Amino acid chain to protein 3D structure.

A substantial majority of bioinformatics study focuses on the estimation, interpretation and simulation of protein 3D structures. Proteins first 3D structure (that of myoglobin) was experimentally determined in 1958, through X-ray diffraction. However, in 1951, Pauling and Corey set the first milestone in the protein structure prediction. As in other fields of biological sciences, it is now possible to predict secondary and tertiary structure using computer calculations and that too with varying degrees of certainty. High-throughput methods have given the knowledge required to relate protein structures to their results. This structural and therapeutic details can be valuable for bioinformatics applications in medical science. Computerized visualization of the protein models provides insights into biological processes that cannot be appropriately described otherwise.

Though advances in 3D structure prediction field are vital, it is significant to know that proteins are a dynamic network of atoms rather than being static. With many advancements in biophysics, force fields have been designed to explain atom interactions between themselves, which enabled the development of tools to model protein molecular dynamics in 1990s. Even though tools were developed and theoretical methods were available, but because of the huge computational resources needed, it remained rather complicated to execute molecular dynamics simulation. It can be explained by the example that it required weeks for the calculation of a microsecond simulation of a protein, using a supercomputer with 256 CPUs. Despite several improvements in the modern computers power like the use of GPU (Graphics processing units) or graphics card, it is still not accessible to perform molecular dynamics simulations on reasonable time scale. But yes, increasing computational power in conjunction with the increasing data have made the process a little bit convenient.

1.7 High-throughput technology

High-throughput sequencing methods have become important in the field of genomic and epigenomic studies. With the advent of increasingly advanced sequencing tools, the amount of DNA Sequencing Approaches has risen tremendously. It has revolutionized the molecular biology field by enabling high-scale whole genome sequencing and also a wide variety of experiments to study the internal cell workings explicitly at the RNA or DNA level. The data generated is the findings of widespread molecular project like gene expression analysis, multiple projects of genome sequencing, protein-protein interactions and analysis of genomics. They are compiled and deposited in a number of databases.

High throughput sequencing is generally divided into two classes: RNA seq and Genome sequencing. In the latter one, sequencing of fragmented genomic DNA is done and reads sequence is used for the assembling of the whole genome. But on the other side, RNA-seq attempts to read the sequence taken from the RNAs. In both the cases, reads may be paired end or single end. For RNA seq, reads are produced from both the ends of the longer fragmented RNA or DNA. While choosing

High throughput technology the user should consider the quality control issues, sample collection, along with the biological hypothesis being tested.

It is a technology developed as an alternative to microarray. Although high throughput technology is also comparatively more costly than the microarray, it still has many benefits for the evaluation of the factors that influences the gene expression regulation. For e.g.: microarray is limited to the model organism to which the microarray has already been built while HTS may be extended to a non-model organism. But High throughput methods are now getting cheaper and are likely to replace even fingerprinting methods including analysis of traditional clone library. High throughput sequencing methods provides the capacity for the detection of rare phylotypes particularly, effectively offering quite reliable estimates of relative abundance and assessment of diversity indices. The key benefit of HTS is that it produces good quality gene expression data sets. There is a need of specialized tool for viewing, storing, indexing, organizing and analyzing biological and computerized data. Thus, bioinformatics is the bridge between computational and biological sciences, which can provide a deeper insight into this field. Data sets of HTS are both complex and high dimensional in nature. It is quite challenging both computationally and algorithmically to integrate such data with various other data sets in order to attain profile of a diseases completely. To integrate HTS data, network -based approaches have the ability to incorporate data from various sources while ensuring results that are relevant. There are many multidisciplinary programs like molecular tumor boards that put together biologist, physicians and bioinformatics, which helps in addressing the challenges of translating the data that are important to health care providers and patients. Substantial computational power is needed by the related algorithmic approaches. High Performance computing (HPC) offers resources which can be exploited by computer/bioinformatics researchers. Some of the resources offered by HPC are cloud computing platforms, GPU (Graphic Processing Unit), or clusters. Every resource is different in terms of performance, technology, ease and scalability of implementation, and cost. The table shows example of High Throughput applications which uses GPU, cloud or other resources ([Table 1.1](#)).

Table 1.1 High throughput computing devices.

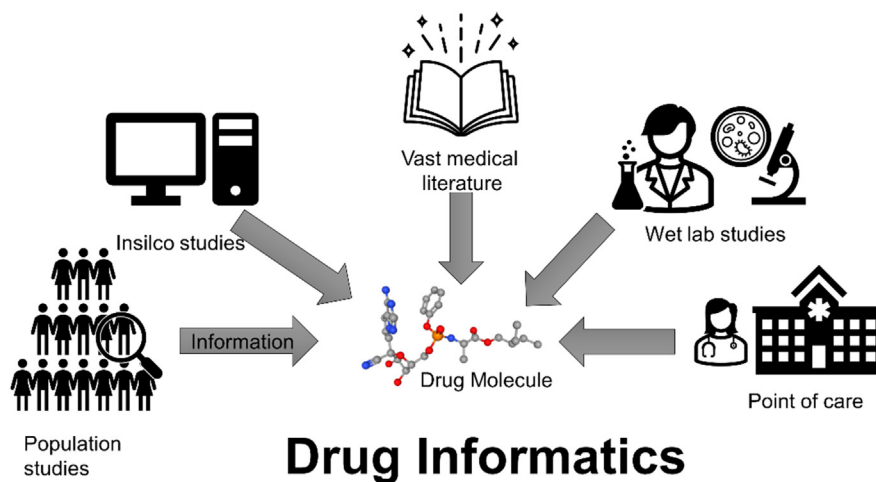
S.NO.	HPC	High throughput applications
1.	GPU	Read mapping, Process-intensive task (RNA-seq alignment)
2.	Clusters	Self-Organizing Maps (SOM) Exome analysis workflow
3.	Cloud	Used by private and public clouds like 1000 genome project and international cancer genome consortium.

1.8 Drug informatics

Drug Informatics relates to the combination of computer techniques and pharmacy expertise to discover and examine drugs. Drug-Informatics is the study of relationships between drugs, their mechanisms, and structures, focusing on medication awareness and improving the quality of life. It is not manually possible for health-care professional to hold all of the information which is required to provide medical care with safety and efficacy aided by the scientific knowledge present today. The scenario can further worsen as there is huge increment in the complexity and volume of data regarding mechanisms of diseases generated through genomic revolution. The solution to this whole mess is the acquiring of thorough technologies and techniques for the management of the data. And that's where drug informatics comes into picture as it enfold the area where these technologies and techniques affect the use of drug data in a commercial, clinical or research setting.

It is the sum of all data and information generated throughout the life cycle of a drug. The delivery of this information from lab to patient is not direct. It involves drug informatics to provide specific information related to the patient. The sources of drug informatics have been classified into 3 categories: primary, secondary and tertiary. Primary level of literature includes data from the research lab of a university or a pharmaceutical company or even the clinical observations from a hospital. It also includes unpublished and published data from various journals but not every published literature in a journal is said to be primary literature, such as editorials or review articles are not said to be considered as primary literature. Secondary level of literature consists of the data created as a repercussion of the use of data originally accumulated for non-clinical purposes like pharmacy benefit reimbursement and management. It also consists of sources that either abstract or index primary literature, with the intention of leading the individual to the appropriate primary literature. Tertiary literature contains material which has been compiled and condensed by the editor or author to provide a short and simple overview of the subject. Few examples of this consist of journal review papers, compendium textbooks and general data which can be downloaded on the internet (Fig. 1.4).

During the evolution of the drug informatics and its practices the healthcare sector was faced significant challenges: rising prices, excessive loss rates, and disgruntled patients and providers. Then entered the information technology applications in the field during the 1960s which focused on clerical and financial systems. With the advent of effective network technology and personal computers in 1980s, there came the introduction of more clinically oriented computing systems for healthcare. It is still a field which is still new particularly in comparison with other disciplines of medical. Drug informatics is an exponentially growing field with the application of information technology and computer science to health and medical data. It encourages the technology use as an essential resource to efficiently manage, analyze, organize data on the use of drugs on patients. The fundamental aim of the field is to disseminate two categories of information knowledge-based information: consists healthcare scientific literature and patient specific information: generated

**FIGURE 1.4**

Schema of drug informatics.

during taking care of the patient. Policy makers and other leaders in health informatics are now inclined toward the improvement of the safety and quality of health care systems while at the same time lowering the cost.

1.9 System and network biology

Network and Systems Biology is the analysis of complex interconnecting systems through the comprehensive method. Network and systems biology attempts to explain the interaction network between hundreds of various biological molecules concurrently. Biological science is undergoing two increasingly evolving phenomenon: first is the increased computer capabilities and advancement in software tools, secondly advent of High throughput biological data on proteomes, transcriptomes, genomes and so on. The biological field has become a data intensive field due to which computer science and biology are now parallel to each other along with fields like algebra, chemistry, physics and statistics. The integration of these fields has contributed in the development of network biology, bigdata and other divisions of biology. The aim of the network biology is to comprehend cells or organisms as a whole at different stages of mechanism and functions. The difficulties of analyzing and evaluating massive biological networks and molecular data is now faced by system biology. The types of data in system biology are sequence data: DNA sequences, Molecular structure data of DNA and RNA extracted by using NMR and X-ray crystallography, Gene expression data, binding sites and domains data, protein-protein interaction and metabolic pathways data. Generation of network is done after receiving data from different data types. After

this multivariate analysis is done on the network, using regression analysis, PCA (Principal Component Analysis) and Clustering. Some of the applications of network algorithm in system biology are: Function prediction: system level approach for determining the functions of an entity which is not known is carried out by network creation of that entity along with the unidentified and identified entities, Protein Complex detection: done using Y2H (yeast two hybrid system) and Affinity purification- Mass Spectrometry, Analyzing evolution, Drug development, Disease diagnosis and Interaction prediction (Crombach and Hogeweg, 2008; van Hoek and Hogeweg, 2006; Lindenmayer, 1968).

1.10 Machine learning in bioinformatics

Machine learning is a technique for analyzing large amounts of data and developing predictive tools. While logical and computational algorithms pass the input data through some logical rules or mathematical operations and provide the desired output, on the other hand machine learning algorithms are trained on data consisting of input and their outputs and formulate the rules for calculating the outputs during training. Thus, machine learning can be used to calculate a predictive output from new input data. Machine learning techniques are based on probabilities and statistical functions, and they attempt to fit the training data into a mathematical model or function. Machine learning has become critical for solving complex problems in the field of genomics. Machine learning has emerged as a savior as the rapid evolution of high throughput technologies outpaces the development of techniques capable of making sense of all available data. As with any other field of science and technology, life science has embraced an Artificial Intelligence-based approach to dealing with the avalanche of data generated (Minsky and Papert, 1969) (Fig. 1.5).

In general, machine learning algorithms can be classified into three categories: supervised learning, unsupervised learning, and reinforcement learning. When both the input and output variables are known, supervised learning is used; this technique is primarily used to develop predictive tools based on known data. Unsupervised learning is used in cases where the output results are unknown, but we attempt to cluster data points based on their similarities or dissimilarities. Unsupervised learning is used to discover correlations and variations in data or to identify outliers. Reinforcement learning is used in situations where a dynamic environment requires multiple outputs or a series of multiple outputs for specific inputs. Algorithms are trained to select the best output based on a reward and punishment system, where the correct output is rewarded and the incorrect output is punished. At the moment, machine learning is being aggressively applied to nearly every problem in bioinformatics. As a result, machine learning has become a necessary tool in the toolbox of every bioinformatician. Chapter 9 of this book contains a more detailed discussion of machine learning algorithms and their applications.

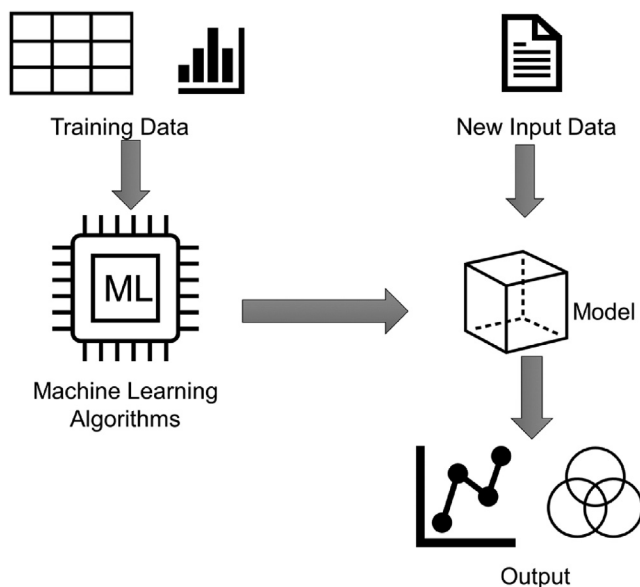


FIGURE 1.5

Schema of a machine learning problem.

1.11 Bioinformatics workflow management systems

A bioinformatics workflow management system is built primarily to compile and implement a sequence of bioinformatics steps. As new pipelines are being developed to analyze the data generated from various sources, standardization of these pipelines becomes essential for the reproducibility of results. Pipelines are first divided into their components then each components are built individually such that their inputs and outputs are in standard form and can be integrated independently when require.

There are numerous workflow frameworks. Experts from various fields like have developed some of the leading research workflow models. Such workflow is based on an abstract description of how a computational pipeline continues in the context of a directed graph. Each framework usually features a visual interface, enabling users to construct complex programs by joining the input and outputs of various components form various pipelines without becoming fluent in computer programming.

Various workflow management systems exist like;

KNIME (Konstanz Information Miner): It is a free, open-source data processing framework. Many tools combine to build complex structures for machine learning and data mining under the modular term “Lego of Analytics.”

Online HPS: OnlineHPC is an online toolset that offers high-performance computing and job flow services to broad agencies. An online group of scholars working on computer systems science has developed.

Galaxy: Galaxy tools help scientists without computer programming to handle computational biology results.

UGENE: UGENE is a software application for bioinformatics. It operates on Windows, Mac OS X, or Linux operating systems. UGENE software helps geneticists examine genomic details such as DNA genomes, genome descriptions, phylogenetic trees, and assembly files. The data can be processed both locally and online (e.g. a lab database).

Gene Pattern: GenePattern is a publicly accessible computational biology open-source software kit produced and built at the Large Institute, built to replicate new genomic analyses by researchers.

1.12 Application of bioinformatics

Bioinformatics is a multidisciplinary area which develops software's tools and methods for analyzing the biological data. The advances in the both computer sciences and molecular biology over the past 30–40 years has contributed in the development of bioinformatics. Bioinformatics key areas include sequence alignment, genomics, proteomics, molecular phylogenetic and many more. It is intended to set up an automated archive that catalogs genomes and protein sequences from single-celled species to multicellular ones. This approach enables three-dimensional structures of complicated molecules to be studied and interpreted. Bioinformatics is a mixture of mathematical, bio-chemical, and numerical approaches to analyze human, biochemical and biophysical evidence. Bioinformatics is concerned with gene sequences and their encoded details.

The bioinformatics field has become important for basic molecular biology and genomic research along with having a significant influence on various fields of biomedical science and biotechnology. Bioinformatics is being used in almost every field of biology starting with medicine the various applications of it are:

Drug Designing: Changes in drug designing happened within a decade when the first 3-D protein structure was illustrated. The information of target protein 3-D structure was incorporated into the drug design procedure. In the process of drug designing the structure of protein can influence the whole process at any stage. Various online tools and software's have been made to make for better accessibility and results. For e.g.: Auto dock.

Personalized medicine: With the advancement of the pharmacogenomics area, clinical medicine can become more personalized. This is an analysis about how genetic inheritance of an individual influences the body reaction to drug. Till now, physicians were using trail and error methods to the determine the right medication to treat a single patient, since people with the same health symptoms can show a wide variety of reactions to the same therapy. In last decade personalized medicine has emerged which allows doctors to prescribe best dosage and available drug therapy from the start by analyzing the genetic profile/biomarkers of the patient ([Anderson et al., 2008](#)).

Gene Therapy: It is the approach used in the curing, preventing and giving treatment of the diseases by changing the expression in the genes of a person. With the use of bioinformatics tools, the possibility for the use of genes itself to cure diseases have become the reality now.

Microbial Genome Applications: The emergence of full genome sequences and their ability to provide deeper understanding of the microbial ecosystem as well as the capabilities, could have far reaching and wide consequences for energy, industrial, environmental, and health applications. After the genetic material analysis of these organisms, researchers may start to know these microbes in their primitive form and identify genes that provides them their remarkable ability to live under harsh conditions.

Evolutionary studies: Genome sequencing of all the three organisms, archaea, bacteria, and eukaryote signify that evolutionary studies could be carried out in the pursuit of determining the tree of life and bioinformatics can surely help using phylogenetics studies.

Computational approaches for bioinformatics expand knowledge and strategies for cellular and molecular levels into information for moving up to the level of inanimate structures such as habitats. Genome sequences offer a way of recognizing the usual biological mechanisms, clarifying the malfunctioning of genes contributing to diagnosing diseases and designing new medicines. There are so many applications of bioinformatics in various fields of agriculture, microbial genome and medicine etc. Using bioinformatics will help biologists and researchers to excel in the experiments and will also helps in extending their skills in data processing more reliably and quickly.

References

- Altschul, S.F., Gish, W., Miller, W., Myers, E.W., Lipman, D.J., 1990. Basic local alignment search tool. *J. Mol. Biol.* 215, 403–410. [https://doi.org/10.1016/S0022-2836\(05\)80360-2](https://doi.org/10.1016/S0022-2836(05)80360-2).
- Anderson, A., Chaplain, M., Rejniak, K., Fozard, J., 2008. Single-Cell-Based Models in Biology and Medicine.
- Burks, C., Fickett, J., Goad, W., 1987. GenBank status report. *Science* 235, 267–268. <https://doi.org/10.1126/SCIENCE.235.4786.267-C>, 1979.
- Chang, M.A., Dayhoff, M.O., Eck, R.v., Sochard, M.R., 1965. Atlas of Protein Sequence and Structure. National Biomedical Research Foundation.
- Crombach, A., Hogeweg, P., 2008. Evolution of evolvability in gene regulatory networks. *PLoS Comput. Biol.* 4.
- Dayhoff, M.O., Barker, W.C., McLaughlin, P.J., 1974. Inferences from protein and nucleic acid sequences: early molecular evolution, divergence of kingdoms and rates of change. *Orig. Life* 5, 311–330.
- Hagen, J.B., 2000. The origins of bioinformatics. *Nat. Rev. Genet.* 1 (3), 231–236. <https://doi.org/10.1038/35042090>.
- Hesper, B., Hogeweg, P., 1970. Bioinformatica: een werkconcept. *Kameleon* 1 (6), 28–29. Dutch, Leiden: Leidse Biologen Club.

- Konings, D., Hogeweg, P., Hesper, B., 1987. Evolution of the primary and secondary structures of the E1a mRNAs of the adenovirus. *Mol. Biol. Evol.* 4.
- Lindenmayer, A., 1968. Mathematical models for cellular interactions in development II. Simple and branching filaments with two-sided inputs. *J. Theor. Biol.* 18.
- Minsky, M., Papert, S., 1969. Perceptrons.
- Schuster, P., Fontana, W., Stadler, P., Hofacker, I., 1994. From sequences to shapes and back: a case study in RNA secondary structures. *Proc. Biol. Sci.* 255.
- Stoletzki, N., 2008. Conflicting selection pressures on synonymous codon use in yeast suggest selection on mRNA secondary structures. *BMC Evol. Biol.* 8.
- van Hoek, M., Hogeweg, P., 2006. In silico evolved lac operons exhibit bistability for artificial inducers, but not for lactose. *Biophys. J.* 91.